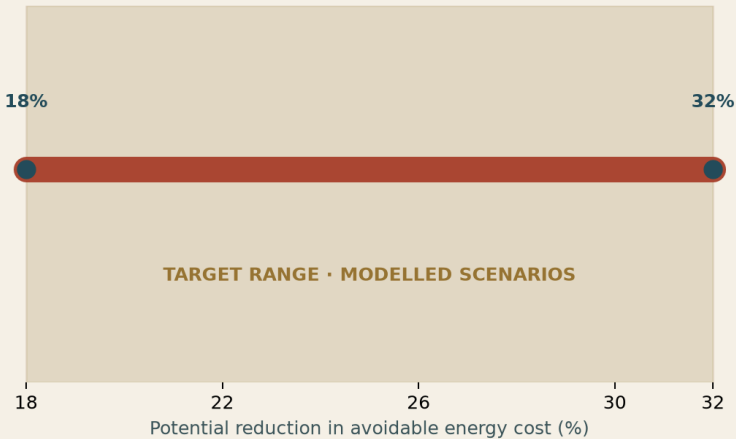


Aura TESLA Opt. Engineering paper

Mission, prediction, operation, scaling and local-first data boundary.

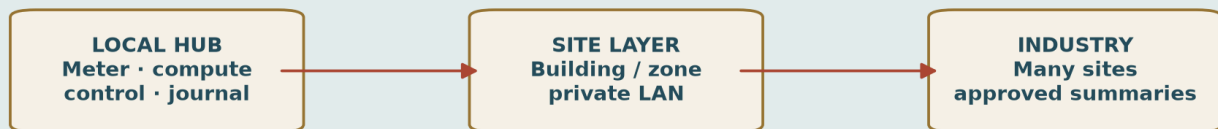


Mission and predicted economics

The system mission is to make consumption observable, shift flexible loads into tariff troughs and coordinate them under deterministic safety policy. The supplied operating model targets **18%–32% lower avoidable energy cost** in suitable scenarios. It assumes tariff spreads, flexible heating and EV load; the range is modelled and must be validated with site data.

Architecture and industrial scale

V5 combines three-phase metering, local compute, four switched channels, Ethernet and a proposed 24 V UPS path. Replication is by adding local hubs per building, zone or production line. Industrial deployment requires segmentation, addressing, fail-safe states, thermal analysis, protection coordination, maintenance and change control.



Scale by adding local hubs and edge devices; keep raw telemetry local when policy allows.

Local-first data boundary

On a private LAN with no cloud relay, raw measurements and control traffic can remain inside the premises. Confirm firewall rules, remote access, retention, credentials, logging, backups and every third-party integration; privacy is a deployment property.

Operational loop

Hourly: observe tariff, prices, rollups, load states and habit digest; propose a structured plan with a local model; validate using deterministic policy; commit decision and command atomically; execute through a retrying outbox. Rejections and failures are visible. Nightly habits are rebuilt in SQL. Weekly feedback is gated before adapter promotion.

Safety boundary

Every switched load keeps its own breaker in the main distribution board. CT clamps face the load and never surround PE. AC installation, protection, insulation, thermal behaviour and final compatibility require a qualified professional.

Status: design and prediction paper. No certification, savings guarantee, battery autonomy or field readiness is claimed.

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